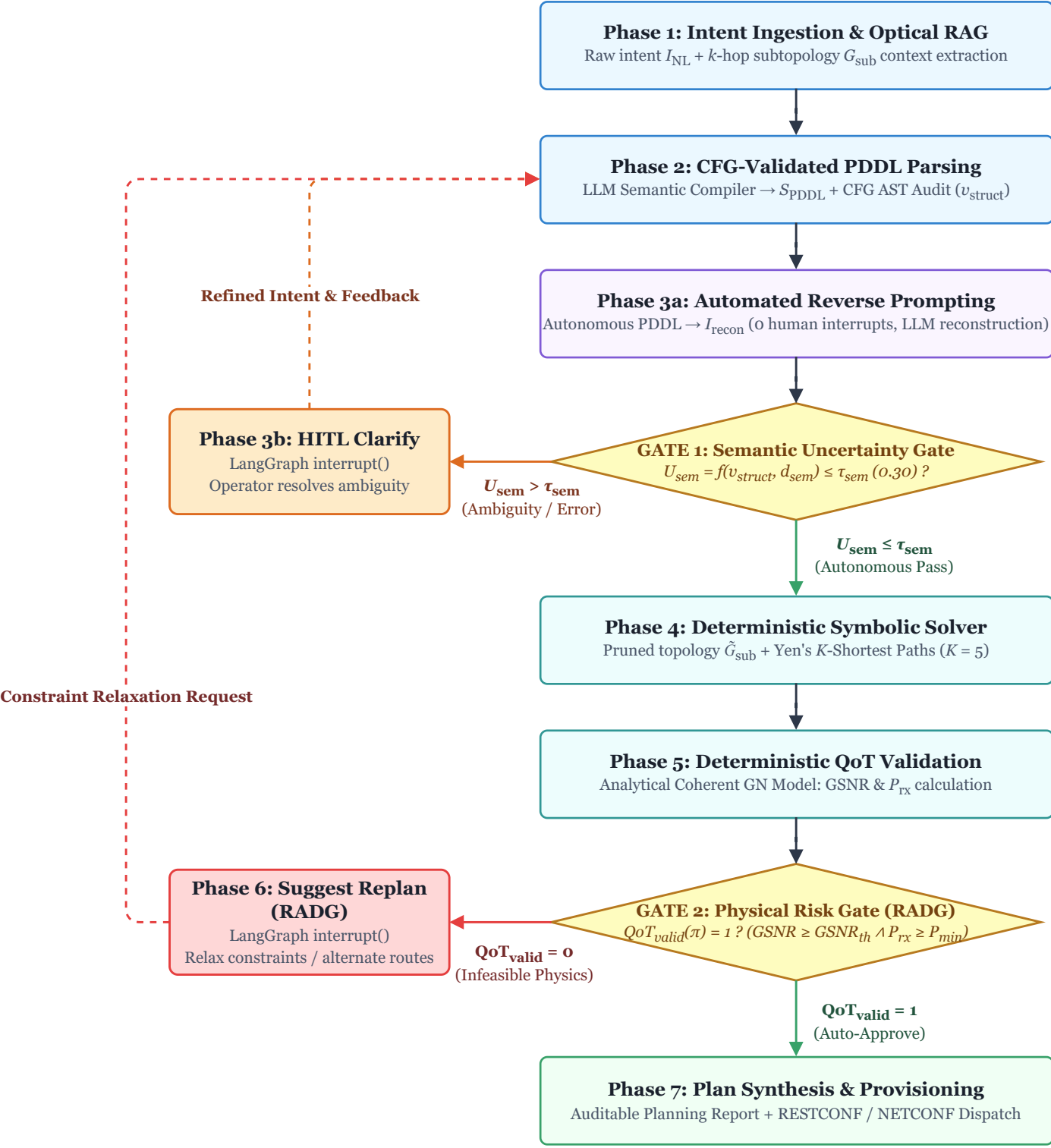




Architectural Invariants & Guarantees

Fail-Fast Hierarchy

Semantic ambiguity resolved in Gate 1 before executing costly topological graph pruning or GN-model physics.

Strict Neurosymbolic Separation

LLM acts solely as a semantic compiler to PDDL. All routing and physical calculations are delegated to deterministic external engines.

Zero Human Friction on Low Risk

If $U_{sem} \leq \tau_{sem}$ and QoT is valid, the pipeline auto-approves with 0 operator interruptions.

Pre-Deployment Safety Guarantee

No configuration reaches the physical network unless $D(U_{sem}, QoT_{valid}) = \text{approve}$, precluding post-deployment failure.

Atomic Checkpoint Pausing

Native LangGraph interrupt() serializes full state tuple. Zero LLM tokens consumed during operator review dwell time.